

# GPU Performance Comparison

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## 1 Introduction

In this note, we document the computational performance of our neural network training workflows across multiple high-performance computing (HPC) facilities and our server. The benchmarking focuses on the execution time consumed during the training process.

## 2 Hardware Specifications

To evaluate computational efficiency, we leverage three distinct computing environments:

- **Our Server:** Equipped with **NVIDIA RTX A6000** graphics cards.
- **NCTS Cluster:** Equipped with the **NVIDIA L40S** and **NVIDIA RTX A6000** graphics cards.
- **ASGC DiCOS:** Equipped with the **NVIDIA L40S**, **NVIDIA RTX 4090**, **NVIDIA A100** and **NVIDIA V100** graphics cards.

## 3 Training Time Comparison

The training runs were executed utilizing identical datasets and unified hyperparameter settings to ensure a fair comparison. The benchmarks track the total training duration across three neural networks: the inclusive signal separation (EW vs. bkg) and two polarization-specific networks (LL vs. TX and LX vs. TT). The results are summarized in Table 1.

Table 1: Benchmark results of the Dense-NN training execution time across different computing facilities.

| Environment  | GPU Hardware     | Total Training Time (hh:mm:ss) |           |           |
|--------------|------------------|--------------------------------|-----------|-----------|
|              |                  | EW vs. bkg                     | LL vs. TX | LX vs. TT |
| Local Server | NVIDIA RTX A6000 | 1:04:31                        | 2:06:52   | 2:13:23   |
| NCTS Cluster | NVIDIA L40S      | 0:19:05                        | 0:33:16   | 0:34:47   |
|              | NVIDIA RTX A6000 | 0:57:09                        | 1:49:39   | 1:52:35   |
| ASGC DiCOS   | NVIDIA L40S      | 0:20:58                        | 0:32:29   | 0:29:03   |
|              | NVIDIA RTX 4090  | 0:20:12                        | 0:33:14   | 0:29:24   |
|              | NVIDIA A100      | 0:34:16                        | 1:05:13   | 0:48:04   |
|              | NVIDIA V100      | 0:36:30                        | 1:04:19   | 0:57:39   |

## 4 Discussion and Observations

### 4.1 GPU Hardware Specifications

To understand the performance discrepancies, the fundamental hardware specifications of GPUs are summarized in Table 2. Crucial metrics such as memory, L2 cache size, and boost clock speed directly dictate the I/O and computational limits for our workload.

Table 2: Hardware specifications of the evaluated NVIDIA GPUs.

| GPU         | Architecture | Memory       | Bandwidth | L2    | Clock    |
|-------------|--------------|--------------|-----------|-------|----------|
| RTX A6000   | Ampere       | 48 GB GDDR6  | 768 GB/s  | 6 MB  | 1.80 GHz |
| V100        | Volta        | 32 GB HBM2   | 900 GB/s  | 6 MB  | 1.53 GHz |
| A100 (80GB) | Ampere       | 80 GB HBM2e  | 1935 GB/s | 40 MB | 1.41 GHz |
| RTX 4090    | Ada Lovelace | 24 GB GDDR6X | 1008 GB/s | 72 MB | 2.52 GHz |
| L40S        | Ada Lovelace | 48 GB GDDR6  | 864 GB/s  | 96 MB | 2.52 GHz |

### 4.2 Decomposition of Execution Time and I/O Analysis

For our lightweight Dense-NN tasks (comprising approximately 5 MB of weights and batch data per iteration), the total execution time is heavily influenced by data movement

rather than pure theoretical TFLOPS. The execution time can be decomposed into a four-stage pipeline:

1. **CPU to VRAM:** Data shuffling, preprocessing, and dispatching via the PCIe bus. Limited by Host CPU frequency.
2. **VRAM to L2 Cache:** Moving data from VRAM (GDDR6/HBM2) into the L2 Cache. Limited by memory bandwidth, latency, and L2 cache capacity.
3. **L2 Cache to Core:** Moving data from L2 cache to the L1 cache/Registers for calculation.
4. **Core Computation:** The actual matrix multiplications in the Tensor Cores. Limited strictly by the GPU’s clock speed.

### 4.3 Key Observations

Based on the benchmark results in Table 1 and the hardware specifications, our observations are as follows:

- **The L2 Cache Advantage and Clock Speed Dominance (L40S & RTX 4090):** The Ada Lovelace cards deliver the best performance, completing the training in nearly one-third the time of the RTX A6000. Due to their massive L2 caches, these GPUs bypass the GDDR6 latency. With data readily available on-chip, the performance is bottlenecked solely by the core computation.
- **Clock Speed Ceiling on High-Bandwidth Flagships (A100 & V100):** The A100 and V100 yield practically identical execution times. Despite the V100 having only 6 MB of L2 cache, its HBM2 memory provides ultra-low physical latency, easily saturating the I/O requirements for this small workload. The A100’s larger L2 cache and extreme HBM2e bandwidth offer no additional benefit here, as the V100 is already not I/O-bound. Consequently, the bottleneck shifts to computation. Since both cards share similar core clock speeds, their training durations naturally tie.
- **Internal I/O Bottleneck and GDDR Latency (RTX A6000):** The RTX A6000 performs the slowest despite having a higher clock speed than the A100. Due to its minimal 6 MB L2 cache and GDDR6 memory, the A6000 suffers from severe latency during Stage 2 (VRAM-to-L2 transfers). The computing cores are frequently starved, waiting for data delivery, rendering its theoretical compute capabilities unachievable.

- **Host CPU Bottlenecks on Identical Hardware (RTX A6000 baseline):** Comparing the RTX A6000 between our Local Server (`pheno-2`) and the NCTS Cluster (`gpuserver03`), NCTS shows a 10–15% speed advantage. This is primarily a Stage 1 bottleneck. The NCTS node utilizes a Xeon Gold 6226R at a higher base clock of 2.90 GHz compared to our local 2.10 GHz CPU. This 38% increase in host CPU frequency accelerates the initial data-feeding and kernel launch processes.